

Energy-Density Calculation + Comparison (Case T2R1NOCEILING)

calc-energy contract formula: $ED = \text{integral}(V \times I_{1C})dt / \text{sum}(\text{layer thickness} \times (1 - \text{porosity}) \times \text{density} \times \text{area})$; electrolyte excluded (parameter set lacks density). Selected design L highlighted in calc.xlsx.

Design	Round	ED Wh/kg	SEI100 nm	SEI500 nm	lowT	anode min V
Baseline Chen2020	-	400.3	449.1	777.9	0.9943	-0.4385
A ThinCC	R2	456.4	449.1	777.9	0.9943	-0.4386
B HiCond (sigma x2, D x1R2t+ 0.4)	R2	404.1	476.8	829.8	0.9945	-0.4494
C SmallNeg (r 3.5 um)	R2	407.0	452.9	771.4	0.9944	-0.4368
D LowLoad (x0.728)	R2	355.6	481.0	836.8	0.9940	-0.4067
E ThickNeg (N/P 1.5)	R3	478.6	494.6	865.6	0.9999	-0.4643
F ThinSep (9 um / 0.55)	R3	458.6	455.0	787.9	0.9943	-0.4348
G ThickNeg+ThinSep	R3	480.0	496.2	868.9	0.9999	-0.4704
H ThickNeg2 (N/P 1.77)	R3	452.1	540.0	979.9	1.0000	-0.5027
I Cool (h 100)	R4	454.1	454.1	786.4	0.9919	-0.4337
J CathPartUp (12.5 um)	R4	376.4	463.3	856.7	0.9999	-0.4455
K TransDown (degenerate)	R4	238.2	68.0	68.0	1.0000	+0.1178
L AnPorUp (SELECTED)	R4	476.7	467.6	811.3	0.9944	-0.3868